

Exercise 1: SQL fundamentals

1. SELECT statement

SELECT *
FROM employees;

ID	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-08-20	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Lauren	Hall	IT	55000	2020-08-10	San Francisco

2. SELECT DISTINCT statement

SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

3. ORDER BY statement

```
SELECT first-name,  
       last-name  
FROM employees  
ORDER BY salary DESC;
```

Result

first-name	last-name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Deniel	Clark
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

4. LIMIT statement

SQL query

```
SELECT *  
FROM employees  
ORDER BY salary DESC  
LIMIT 5;
```

Results

id	first_name	last_name	department	salary
6	Dave Emily	Davis	IT	62000
3	Mike	Johnson	finance	60000
7	Robert	Wilson	finance	59000
1	John	Doe	IT	55000
4	Sarah	Brown	IT	53000

5. WHERE statement

SQL query

SELECT *

FROM employees

WHERE department = 'IT';

Results

id	first_name	last_name	department	salary	city
1	John	Doe	IT	55000	New York
4	Sarah	Brown	IT	53000	New York
6	Emily	Davis	IT	62000	Chicago
10	Laura	Hall	IT	50000	San Francisco

6. AND statement

SELECT *

FROM employees

WHERE department = 'finance'

AND salary > 5500;

id	first_name	last_name	department	salary
3	Mike	Jordan	finance	60000
7	Robert	Wilson	finance	54000

7. OR statement

SELECT *

FROM employees

WHERE department = 'HR'

OR department = 'marketing';

id	first_name	last_name	department
2	Jane	Smith	HR
5	Daniel	White	Marketing
8	Jessica	Moore	HR
9	Daniel	Clarke	Marketing

5. NOT statement

SELECT *

FROM employees

WHERE NOT department = 'IT' ;

Results

ID	first_name	last_name	department
2	Jane	Smith	HR
3	Mike	Johnson	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moon	HR
9	Daniel	Clamp	Marketing

6. IN statement

SELECT *

FROM employees

WHERE department IN ('HR', 'IT', 'Finance') ;

ID	first_name	last_name	department
1	John	Doe	IT
2	Sam	Smith	HR
3	Mike	Johnson	Finance
4	Seal	Brown	IT
8	Emily	Davis	IT
6	Robert	Wilson	Finance
9	Jessica Mason	Moon	HR
10	Sam	Hall	IT

10. Combining conditions

SELECT X

FROM employees

WHERE department = 'IT'

AND salary > 50000

AND city = 'New York';

ID	first_name	last_name	department	salary	city
1	John	Dee	IT	\$5000	New York
4	Sarah	Brown	IT	\$3000	New York

11. Where + and + order By

SELECT first_name,

last_name

FROM employees

WHERE department IN ('Finance', 'Marketing')

AND salary > 52000

ORDER BY salary DESC;

first_name	last_name
Mike	Johnson
Robert	Williams
Daniel	Chang

12. DISTINCT + WHERE + IN

SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');

city
Los Angeles
San Francisco
Houston
Chicago

13. WHERE + NOT + AND + ORDER BY

SELECT *

FROM employee

WHERE department ~~is~~ (finance)

AND salary > 50000

ORDER BY hire_date ASC

id	first_name	last_name	department	salary	hire_date
6	Emily	Davis	IT	60000	2015-02-10
5	David	White	Marketing	52000	2016-04-10
5	Jessica	Moon	HR	51000	2015-05-22
1	John	Brown	IT	58000	2014-06-15
4	Susan	Brown	IT	53000	2021-03-28
9	Daniel	Clark	Marketing	53000	2022-06-01

14. KUTHERE + OR + IN * LIMIT

SELECT *

FROM employees
~~WHERE~~ city IN ('chicago', 'los Angeles')
AND department IN ('IT', 'Marketing')

id	first-name	last name	department	City
6	emily	Dani	IT	chicago
9	Daniel	clark	Marketing	chicago